

Michael Penhallegon

Data Platform Engineer | Software Engineer
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SUMMARY

Data platform engineer with 8+ years designing and operating production data systems across biomedical research and enterprise infrastructure. Specializes in ELT pipelines and lakehouse architectures built on distributed query engines and warehouse-native transformation, with data quality and observability engineered into the pipeline rather than added after the fact. Applies an API-first approach to integration and incorporates agentic, LLM-assisted development into the engineering lifecycle under disciplined governance.

- 8+ years of production Python and SQL across distributed data systems, with working R and Scala
- ELT and lakehouse architecture: PySpark, Delta Lake and DuckDB, Apache Airflow orchestration, AWS Athena, and warehouse-native transformation
- Data quality and observability engineered into the pipeline: layered testing with pytest and Great Expectations, Prometheus and Grafana instrumentation, and payload-level integrity checks
- API-first integration: FastAPI microservices, REST patterns, and internal client libraries that insulate consumers from backing-store change
- Container-native delivery and AI-assisted SDLC: Docker and Podman, CI/CD, and governed LLM-augmented development

EXPERIENCE

Hematite Solutions, Remote

November 2025-Present

Founder and Principal Engineer

- Founded an independent software and data consulting practice delivering web and edge-architecture solutions across WordPress, JAMstack, and Cloudflare Workers, Pages, and R2
- Built TypeScript edge applications with CI/CD and automated testing (GitHub Actions, Vitest, Miniflare)
- Brought agentic, AI-assisted development into the SDLC under explicit governance

Crown Castle, Seattle, WA

April 2023-April 2025

Software Engineer II

- Built data pipelines and dashboards exposing real-time IP utilization metrics, used by planning and procurement teams to drive allocation decisions and cut forward-looking IPv4 requests to ARIN by 20%
- Developed and maintained Python microservices (gRPC and REST) for surfacing, analyzing, and integrating data from telecom device inventories and other systems
- Developed and maintained PySpark data pipelines
- Led integration testing and expanded unit test coverage, reducing regression incidents across releases
- Integrated monitoring, alerting, and dashboards
- Wrote product plans and project documentation to support team planning

DomainTools, Seattle, WA

July 2021-August 2022

Software Engineer II

- Developed and maintained Python FastAPI microservices
- Applied unit and integration testing to maintain code quality
- Integrated and configured Prometheus alerting and Grafana dashboards
- Traced bugs and data quality issues from front-end through back-end and storage systems
- Wrote complex PostgreSQL functions and stored procedures over large datasets to transform data and maintain warehouse metadata
- Built integrated reporting across multiple microservices and databases, delivering operational analytics to stakeholders in a hybrid on-premise and cloud environment (AWS S3, Lambda)

Bristol Myers Squibb, Seattle, WA

August 2018-June 2021

Software Engineer

- Hired by Juno Therapeutics; the company was acquired by Celgene, then Bristol Myers Squibb, during tenure
- Built and maintained data infrastructure for CAR-T cell therapy translational research, supporting clinical and biologics data workflows
- Automated data ingestion and delivery, cutting transfer times from days to hours, with cleaned data landed in S3 and queried through AWS Athena
- Led requirements gathering and project planning with research partners to deliver data, software, and visualization tools
- Built visualizations, web applications, and REST APIs in R and Python
- Supported and implemented the migration of the team's analytical source of truth from a Postgres-backed LabKey research platform to an AWS Redshift warehouse during enterprise consolidation, building and iterating the ingestion pipelines (Airflow, containerized DAGs) and the internal client libraries that kept downstream consumers stable as the backing store changed
- Built ingestion-side integrity controls including payload hashing for duplicate detection, preventing data corruption during the dual-run period when both the legacy and new source-of-truth systems were live
- Built five-plus internal packages and libraries giving the team consistent, reusable access to upstream data and REST APIs

Fred Hutchinson Cancer Research Center, Seattle, WA

June 2017-June 2018

Research Intern

- Placed in the HIV Vaccine Trials Network Core Lab across two internship terms (Summer 2017 and Spring 2018), with continuing work between terms
- Developed and published a CRAN R package to linearly interpolate Census data into health-metric values
- Performed data analysis and prepared results for academic publication

TECHNICAL SKILLS

Languages: Python, SQL, R, Scala, TypeScript

Data and Processing: PySpark, Apache Spark, Delta Lake, DuckDB, DataFusion, Apache Airflow, ELT, lakehouse architecture, data pipelines

Cloud and Infrastructure: AWS (S3, Athena, Lambda, Redshift), Cloudflare Workers, Docker, Podman, container orchestration, CI/CD, GitHub Actions

Databases: PostgreSQL, MongoDB, Redis

Quality and Observability: pytest, Great Expectations, Prometheus, Grafana

Practices: API design (FastAPI, REST), data quality, data governance, agile and scrum, AI-assisted development

EDUCATION

The Evergreen State College, Olympia, WA

June 2018

Bachelor of Arts, emphasis in Computer Science and Public Health Studies

VOLUNTEER AND LEADERSHIP EXPERIENCE

Sisters of Perpetual Indulgence, Abbey of St. Joan, Seattle, WA

September 2019-Present

Member

- Held progressive governance roles: Secretary (2021-2023), Board Member (2023-2024), and President (2024)
- Led digital communications and outreach for a community health and social-justice nonprofit
- Coordinated public-facing communications for community fundraising events

Additional Leadership and Service

- King-Snohomish-Island Counties Ryan White RFP Review Committee, Member (2021): reviewed and scored grant proposals for Ryan White HIV/AIDS Program funding allocations